

Analyse and connect a variety of data and information to identify and explain patterns, trends, relationships and anomalies

Learning intention

- analyse and connect a variety of data and information to identify and explain patterns, trends, relationships and anomalies.

Teach and model

- discussing the validity of the data when extrapolating from a graph.
- examining tables, graphs and digital simulations of radioactive decay half-life to predict changes in mass over time.
- analysing data on heat transfer through multiple layers of an insulating material and identifying patterns and proportional relationships.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.